Fiscal Year:	FY 2026	**Task Last Updated:**	FY 12/30/2025
PI Name:	Hofmann, Douglas Ph.D.		
Project Title:	Study of Mushy-Zone Development in Dendritic Microstructures with Glass-Forming Eutectic Matrices		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 9/30/2025 per PI (Ed., 8/7/24) NOTE: Extended to 6/30/2023 per PI (Ed., 8/5/21) NOTE: Extended to 9/30/2021 per PI (Ed., 7/8/19) NOTE: Extended to 9/30/2019 per PI (Ed., 7/5/16) NOTE: Extended to 10/1/16 (Ed., 7/31/13) NOTE: To be extended past 10/1/2013 per PI (Ed., 7/24/13)		
Key Personnel Changes/Previous PI:	Dr. Punnathat Bordeenithikasem is no longer a Co-Investigator on the project due to a reduction in workforce. Postdoctoral scholar, Dr. Miguel de Brito Costa contributed to the program.		
COI Name (Institution):	Johnson, William Ph.D. (California Institute of Technology) Roberts, Scott Ph.D. (Jet Propulsion Laboratory)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	The future materials for NASA spaceflight applications are likely to be high-performance composites. Metallic glass matrix composites, which are metallic glasses reinforced with crystalline dendritic inclusions, were found to impart benchmark properties (see for example, D.C. Hofmann et al. Nature, 451:1085, 2008). The mechanical performance of these composites is understood to be highly dependent on the morphological-feature length scales of the dendritic microstructure. Specifically, it has been demonstrated that the length scales associated with the size and distribution of dendrites has a profound effect on mechanical properties such as ductility, fracture toughness, and fatigue endurance. Understanding and controlling the development of the dendrite microstructure will enable optimization of the mechanical properties and promote a truly unique mechanical performance of these composites. Amorphous metals (AMs), also known as Bulk Metallic Glasses (BMGs), are a unique class of metal alloys known for their non-crystalline microstructures, unique mechanical properties, and low melting temperatures. BMG matrix composites are amorphous metals reinforced with crystalline phases to improve mechanical properties such as toughness, strength, and wear resistance. When BMGs become crystalline, their toughness and wear resistance is normally severely decreased. By strategically adding crystalline particles, such as dense tungsten (W), within the crystalline BMG matrix, the wear resistance can be remarkably improved. However, sedimentation dramatically affects how these composites can be processed. The microgravity environment aboard the International Space Station (ISS) will be utilized to perform a semi-solid, isothermal (or mushy) hold on the composites between the solidus and the liquidus to selectively precipitate out dendrites into the BMG matrix, resulting in alloys that have wear resistance and toughness that cannot be matched through terrestrial processing.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of innovative new materials is a fundamental part of future terrestrial and space technologies. These materials have a tremendous benefit towards improving life on Earth, from improving the accessibility and cost of solar power, to making cars safer and stronger, to reducing the mass of aircraft, for example. Innovative technologies typically require new materials tailored for those applications. This drives materials development. Metal matrix composites (MMCs) are a relatively new class of metal alloys that exhibit unique properties which make them desirable for many new technologies. MMCs are typically two-phase alloys consisting of a continuous metal matrix incorporated with a second phase (which can be a metal dendrite, a fiber, a particle, or some other inclusion). They differ from traditional monolithic crystalline metal alloys because the mechanical and physical properties of single-phase alloys are limited by composition and dislocation-based plasticity. By creating a composite phase, MMCs have the ability to combine the beneficial properties of both phases into a single alloy, like strength and toughness, for example. A novel type of MMC has been recently developed that exhibits benchmark combinations of strength and toughness. These alloys were developed by creating a MMC with high-strength metallic glasses (also called amorphous metals) as the matrix phase and a soft body centered cubic (bcc) dendrite as the crystalline phase. These metallic glass matrix composites (MGMCs), which are the focus of the current research effort, have been shown to have the strength of carbides, with the toughness and fatigue limit of stainless steel and the density of titanium. These alloys have the potential for significant impact in high-performance structural applications in areas ranging from defense, to aerospace, to commercial sporting goods. The widespread use of MMC (and MGMCs in particular) requires in-depth research into the processing and fabrication of these multi-phase alloys. For example, semi-solid (or mushy) processing is often required to coarsen the second phases in MMCs and MGMCs to improve the mechanical properties. These additional processing requirements make the fabrication of hardware from MMCs unique compared with traditional metal parts (which are typically machined into a final shape). The NASA-funded technology being developed as part of the current program seeks to understand the development of second phases in MGMCs as a function of processing parameters so that innovative industrial processing techniques can be used to generate hardware. The end result is the potential widespread use of MMC in applications such as biomedical implants or spacecraft shielding.
	The FAMIS experiment demonstrated several results that will have impact on future alloying and manufacturing in microgravity. The primary result was that processing the two glass-forming alloys in microgravity, when embedded with inclusions with a significant density difference between the alloys, resulted in unique microstructures that could not be obtained on the ground due to sedimentation. Therefore, processing in microgravity produced novel materials that could have properties that cannot be replicated in 1g processing, in the case of FAMIS, alloys with extremely low wear-loss due to the homogenous phases produced during microgravity processing. The FAMIS experiment was also successful at determining the sensitivity of the alloys to changes in the diameter and volume fraction of the dense W inclusions as well as the effect of two different glass-forming alloys. It was determined that the Vit106 alloy became extremely brittle when cooled slowly from 1200 °C and its wear resistance was significantly improved by the presence of W particles and improved even further by the presence of small, homogeneously distributed W2Zr particles formed by reacting the W with the Zr in the matrix. In both alloys, Vit106 and CuZrAlCo, the W reacted only with the Zr in the matrix to form the W2Zr intermetallic phase, which significantly changed the composition of the CuZrAlCo alloy because the primary constituent of the alloy was Cu, not Zr. This created a copper-rich matrix alloy that was far away from the original glass-forming alloy. This was one of the most unique observations from the FAMIS experiment. By holding the two alloys in the semi-solid region and allowing them to react with W, the matrix of each alloy was depleted in just Zr, which was used to create an entirely new intermetallic phase. With this knowledge, the initial composition of both glass-forming alloys should have been overalloyed with Zr such that after processing, the composition of the remaining matrix was at the optimal glass-forming composition, after being depleted in Zr to form the intermetallic phase. The FAMIS experiment also demonstrated that the rate of W diffusion to form the intermetallic phase could be controlled with the diameter of the initial W particles. The large W particles did not completely dissolve in 120 minutes at 1200 °C, however, the smaller W particles completely dissolved with the same volume fraction and processing conditions. The FAMIS experiment therefore demonstrated in-situ alloying that was used to create entirely new compositions and microstructures; either composites comprised of W2Zr in a depleted Zr matrix or composites comprised of both W2Zr and W in a depleted Zr matrix. Next, the FAMIS experiment demonstrated that the effect of sedimentation could be reduced or eliminated with alloy selection. The ground-based experiments showed that with sufficient volume fraction of

Task Progress:	small W particles, sedimentation under 1g processing could be completely avoided, even with a significant density difference between the matrix alloy and the W-based phases. This happens due to the rapid diffusion of the smaller W particles and a resulting scaffolding effect that prevents sedimentation. This demonstrates that some metal alloys and composites have sensitivity to being processed in 1g due to sedimentation, whereas others do not, based on factors such as viscosity, phase formation, and diffusion rate. One of the deleterious observations from the FAMIS experiment was the extent of sedimentation observed in the samples due to the absence of gravity during processing on the ISS. The observed porosity took two forms; (1) large, coalesced porosity in the interior of the sample and (2) microporosity formed from cavitation during solidification. The observed porosity varied between the two glass-forming alloys tested, with Vit106 having larger, coalesced porosity, and CuZrAlCo having more homogeneous, cavitation-induced porosity. This is likely due to the viscosity differences in the alloys at the processing temperature, where in the lower viscosity alloy, Vit106, the mobility barrier for porosity is lower, allowing bubbles to coalesce. Without the sedimentation forces provided by gravity, the porosity observed in all the microgravity samples was much more severe than what was observed in the cast samples or in the samples processed at 1g. In the as-cast samples, the only observed porosity was casting porosity from rapid solidification. In the samples processed at 1g, porosity originating from gaps between the sample and the wall of the crucible tended to float to the surface of the sample, leaving the rest of the sample nearly porosity-free. The effect of gravity also prevented much of the cavitation-induced porosity from forming, since the liquid had a free surface to contract and relieve the negative pressure that induces porosity. However, in microgravity, there is no buoyancy to drive the porosity to the top of the sample, and thus, the trapped porosity formed when the sample wets the crucible wall, is trapped in the middle of the sample. Moreover, since the sample completely wets the entire interior of the crucible wall due to capillary forces, there was significant negative pressure during solidification, which created many small pores distributed homogeneously throughout the solidified sample. In both alloys, the extent of porosity made mechanical testing challenging, as careful consideration had to be taken to avoid pores during hardness testing and wear-testing. The extent of porosity observed in the FAMIS samples should be considered in future experiments involving metal alloy casting, particularly related to the sample wetting the mold and solidifying without a back pressure. The FAMIS samples, which can be considered to have been cast inside W crucibles, showed extensive porosity due to the wetting of the samples against the W wall and the lack of a compressive force on the liquid during solidification. For future casting or injection-molding experiments on the ISS, a piston or plunger is needed to force the liquid into contact with the mold and to drive out the porosity. Careful consideration must be paid to the design and control of the porosity using strategically located vents to remove the trapped porosity.
Bibliography Type:	Description: (Last Updated: 01/01/2026)
Articles in Peer-reviewed Journals	B. Costa M, Vecchio KS, Kozachkov H, Hofmann DC. "Alloy development of ductile and strain-hardening Fe–Ni-based metallic glass composite." Adv. Eng. Materials. 2025 Dec 7:e202501931. https://doi.org/10.1002/adem.202501931 , Dec-2025